

Lecture 7: Reinforcement Learning

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(based on a slideset by Jelle Zuidema & Iris Proff)

Machine Learning and Language Models, 2026

What is Reinforcement Learning?

What is Reinforcement Learning?

Q-learning

RL in the brain

Model-based RL

Extensions of Q-learning

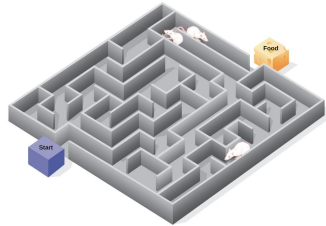
Deep RL

Reinforcement learning

How does an agent learn which action is best in a given situation...

- in an **uncertain** environment
- without a **teacher** signal
- but in presence of (delayed) **feedback?**

RL means learning by interacting with the environment



Types of learning algorithms

- Supervised learning
Learning from labeled examples – neural networks
- Unsupervised learning
Findings patterns in the raw data - clustering
- Semi-supervised learning
Learning from reward signals – Reinforcement learning

Reinforcement learning

A structural model from behavioral psychology...

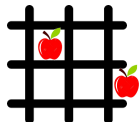
Pavlov: 1897 Pavlovian learning / classical conditioning
→ agent learns stimulus-outcome contingency



Thorndike: 1905 Law of effect / instrumental conditioning
→ agent learns action-outcome contingency



Tolman: 1930 Latent learning / goal directed learning
→ agent learns cognitive map



Reinforcement learning



Reinforcement
Learning
An Introduction
second edition
Richard S. Sutton and Andrew G. Barto

...formalized by AI to a cognitive model

Structural model

Classical conditioning

Cognitive model

→ Rescorla-Wagner model (1972)

Instrumental conditioning → Temporal difference learning (Samuel 1959)

→ Q-Learning (Watkins & Dayan 1992)

Goal directed learning → Model-based learning

Determining utility (optimal control theory)



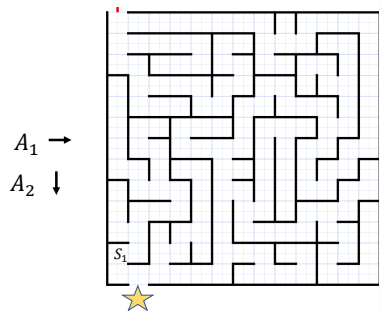
$$U(S) = R(S)$$

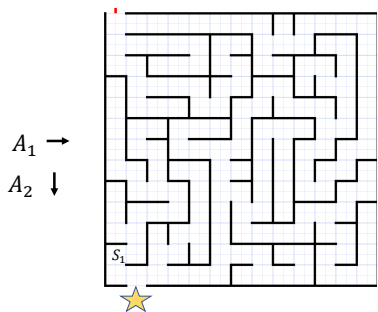


R_1



R_2



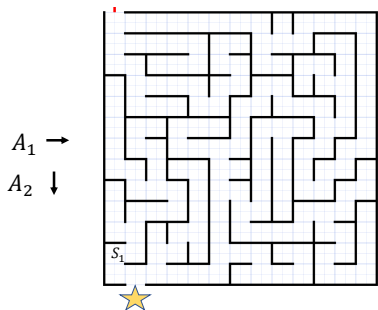


Question 1

Consider states S_2 and S_3 (which can you reach with action A_1 and A_2) and state S_4 (which you can reach from state S_3 by action A_1). How valuable are each of these states relative to each other?

Is $U(S_1) = U(S_3)$?

Determining utility (optimal control theory)



Bellman equation:

$$U(S) = R(S) + \gamma * \max_A [U(S')]$$

U: utility

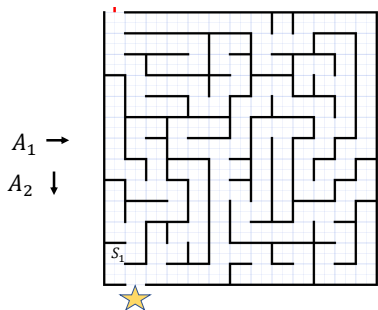
S: state

R: reward

A: action

γ : temporal discount factor

Determining utility (optimal control theory)



Bellman equation:

$$U(S) = R(S) + \gamma * \max_A [U(S')]$$

U: utility

S: state

R: reward

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Question 2

How do you avoid having to first learn everything about the environment/problem before you can start acting in the world?

Q-learning

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Model-based RL

Extensions of Q-learning

Deep RL

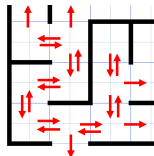
Q-Learning / temporal-difference learning

How to learn the utility of state-action pairs by trial-and-error?

Learning the Q-value = approximation of utility

$$Q(S, A) = \hat{U}(S, A)$$

Online learning (policy is applied and learned at the same time)



Q-Learning – update rule

Prediction error

$$\delta = R + \gamma * \max_{A'}(Q_t(S', A')) - Q_t(S, A)$$

Q-value update

$$Q_{t+1}(S, A) = Q_t(S, A) + \alpha * \delta$$

Q-value approximates utility

$$\lim_{t \rightarrow \infty} Q_t(S, A) = U(S, A)$$

Reminder: Bellman equation

$$U(S, A) = R(S, A) + \gamma * \max_{A'} [U(S', A')]$$

S: state

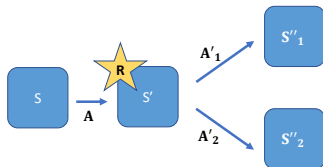
R: reward

A: action

γ : temporal discount factor

α : learning rate

U: utility



Q-learning – action selection policy

Say we are in state S , with possible actions A_1 and A_2 , with
 $Q(S, A_1) = 0.5$ and $Q(S, A_2) = 0.7$

Which action should we choose?

Q-learning – action selection policy

Say we are in state S , with possible actions A_1 and A_2 , with
 $Q(S, A_1) = 0.5$ and $Q(S, A_2) = 0.7$

Which action should we choose?

Question 3

How do you make sure you maximize reward, both in the short term (taking advantage of your knowledge) and in the long term (extending your knowledge)?

Exploration – exploitation trade-off



exploit what you know

should you...

or

explore new options?



“A reinforcement learning agent has to exploit what it has already experienced in order to obtain reward, but it also has to explore in order to make better action selections in the future.

The dilemma is that neither exploration nor exploitation can be pursued exclusively without failing at the task.”

Sutton & Barto 2018

Exploration – exploitation trade-off



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Sutton & Barto 2018

Classic example: [multi-armed bandit](#)

Multi-armed bandit



The name comes from imagining a gambler at a row of slot machines ("one-armed bandits"), who has to decide which machines to play, how many times to play each machine and in which order to play them, and whether to continue with the current machine or try a different machine.

Q-learning – action selection policy

Say we are in state S , with possible actions A_1 and A_2 , with
 $Q(S, A_1) = 0.5$ and $Q(S, A_2) = 0.7$

Which action should we choose, to trade off exploration and exploitation?

Epsilon greedy: $P(a_i|S) = \begin{cases} \frac{1-\epsilon}{|max_{q_i}(A)|} & \text{for } q_i = max(A) \\ \frac{\epsilon}{|A|-|max_{q_i}(A)|} & \text{else} \end{cases}$ best action chosen with $P = 1 - \epsilon$

Softmax: $P(a_i|S) = \frac{e^{\frac{q_i}{\tau}}}{\sum_A e^{\frac{q_i}{\tau}}}$ for $\tau \rightarrow 0$ greedy, for $\tau \rightarrow \infty$ random

Q-Learning – pseudo code

Set $\alpha \in (0,1]$, small $\varepsilon > 0$, $\gamma > 0$

Initialize $Q(s, a)$ for all $s \in S, a \in A$ to 0

Initialize s

For number of episodes

While s is not a terminal state

choose a from s using action selection policy

observe R and s'

update Q

update s to s'

RL in the brain

What is Reinforcement Learning?

Q-learning

RL in the brain

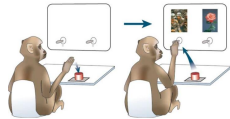
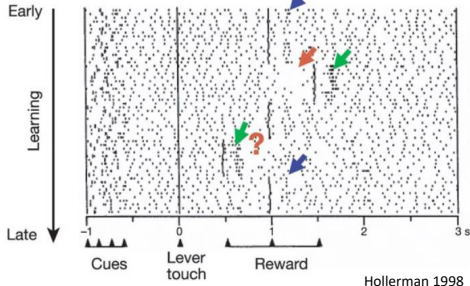
Model-based RL

Extensions of Q-learning

Deep RL

Temporal difference learning in the brain

Recording from a single dopaminergic neuron in monkey brain



Upper blue: expected reward
Red: expected reward omitted
Green: unexpected early/late reward
Lower blue: expected reward

Model-based RL

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RL in the brain

Model-based RL

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Model-based vs model-free learning

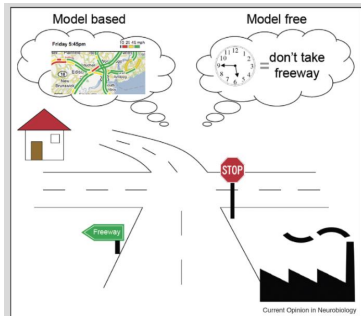


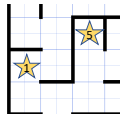
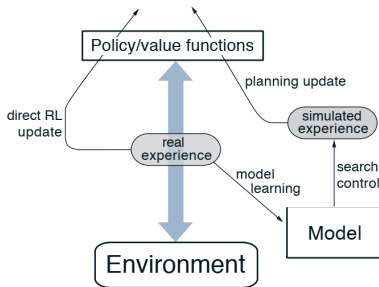
Figure 1: Two ways to choose which route to take when traveling home from work on Friday evening.

Model-free (habitual): rely on trial-and-error, efficient but inflexible

Model-based (goal directed): build a model of the environment, effortful but flexible

Dayan & Niv 2008

Model-based reinforcement learning



Model: $(s, a) \rightarrow (s', r)$

»What does my environment look like? «

Sutton & Barto 2018

Dyna-Q – pseudo code

```
initialize parameters
for number of episodes
  while s is not a terminal state
    choose a from s using action selection policy
    observe R and s', update Q, update s
    update model  $M(s,a) = (R,s')$ 
    repeat n times
      choose (s,a) to simulate
      simulate action: observe R and s'
      update Q
```

direct RL update

real experience

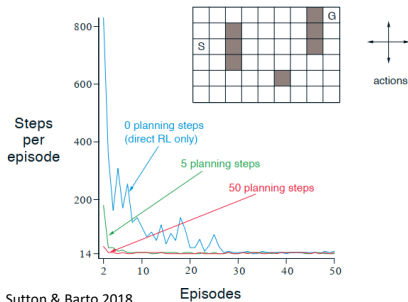
model learning

search control

planning update

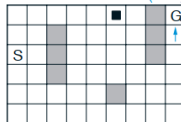
planning

Model-based learning *Dyna-Q*

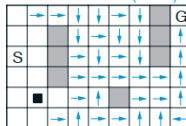


policy during episode 2

WITHOUT PLANNING ($n=0$)



WITH PLANNING ($n=50$)



Extensions of Q-learning

What is Reinforcement Learning?

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RL in the brain

Model-based RL

Extensions of Q-learning

Deep RL

Extensions of RL algorithms

- Eligibility traces

update not only last state-action pairs, but all previously visited state-action pairs with decay factor λ

- Decaying learning rate

α decreases over time

- Hybrid models

Model-based and model-free component

...

Deep RL

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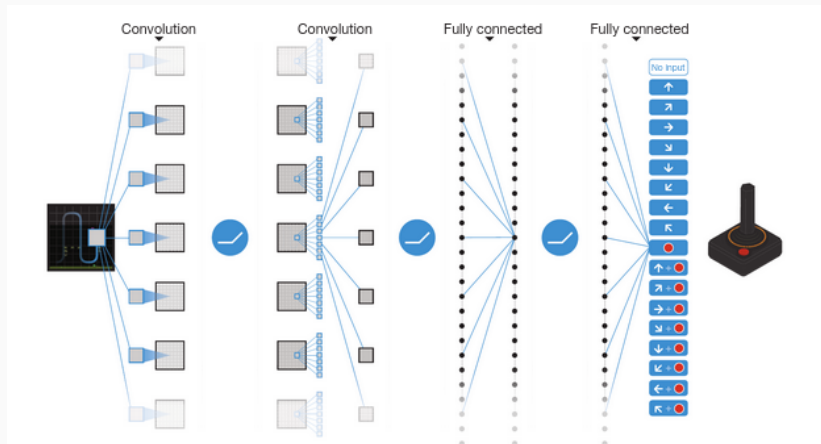
RL in the brain

Model-based RL

Extensions of Q-learning

Deep RL

Deep Reinforcement Learning



Deep Q-Learning (Mnih et al. 2015)

- Replace Q-table with a neural network Q , with weights θ ; use it to compute Q-value of current state
- Make a copy of the Q-network (the *target-network* $\hat{Q}$); use it to compute Q-value of target states
- Learn in batches: act in the world (choose actions using ϵ -greedy) and gather data
- Optimize Q-network (minimize prediction error) via gradient descent on a batch of observations
- Occasionally copy Q into $\hat{Q}$

Deep Q-Learning (Mnih et al. 2015)

Algorithm 1: deep Q-learning with experience replay.

Initialize replay memory D to capacity N

Initialize action-value function Q with random weights θ

Initialize target action-value function $\hat{Q}$ with weights $\theta^- = \theta$

For episode = 1, M **do**

 Initialize sequence $s_1 = \{x_1\}$ and preprocessed sequence $\phi_1 = \phi(s_1)$

For $t = 1, T$ **do**

 With probability ϵ select a random action a_t

 otherwise select $a_t = \operatorname{argmax}_a Q(\phi(s_t), a; \theta)$

 Execute action a_t in emulator and observe reward r_t and image x_{t+1}

 Set $s_{t+1} = s_t, a_t, x_{t+1}$ and preprocess $\phi_{t+1} = \phi(s_{t+1})$

 Store transition $(\phi_t, a_t, r_t, \phi_{t+1})$ in D

 Sample random minibatch of transitions $(\phi_j, a_j, r_j, \phi_{j+1})$ from D

 Set $y_j = \begin{cases} r_j & \text{if episode terminates at step } j+1 \\ r_j + \gamma \max_{a'} \hat{Q}(\phi_{j+1}, a'; \theta^-) & \text{otherwise} \end{cases}$

 Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ with respect to the network parameters θ

 Every C steps reset $\hat{Q} = Q$

End For

End For

LETTER

Human-level control through deep reinforcement learning

Vadavyanet Mhri¹, Koray Kavukcuoglu^{1*}, David Silver², Andrej A. Rusu¹, Joel Veness¹, Marc G. Bellemare¹, Alex Graves¹, Martin Schiedt¹, Andrew N. Senior¹, Greg Deucourt¹, Charles Beattie¹, Iain Durrant-Whyte¹, Demis Hassabis¹, Helen King¹, Olan Shum Khanna¹, Duan Wierstra¹, Shane Legg¹ & Demis Hassabis¹

The theory of reinforcement learning provides a normative account, deeply rooted in psychological and neuroscientific perspectives on animal behaviour, of how agents may optimize their control of an environment. To reuse reinforcement learning successfully in situations approaching real-world complexity, however, agents are confronted with a difficult task: they must derive efficient representations of the environment from high-dimensional sensory inputs, and use these representations to predict the value of future actions. Remarkably, humans and other animals seem to solve this problem through a combination of reinforcement learning and hierarchical or goal-directed processing systems¹. The human evidence is a wealth of neural data revealing parallel parallel between the phasic signals emitted by dopaminergic neurons and temporal difference reinforcement learning algorithms². While reinforcement learning agents have achieved some success in a variety of domains³, their applicability has previously been limited to domains in which useful features can be hand-engineered, or to domains with fully observed, low-dimensional state spaces. Here we use recent advances in training deep neural networks⁴ to develop a novel artificial agent for learning a Q-network, that can learn successful policies directly from high-dimensional sensory inputs using end-to-end reinforcement learning. We tested this agent on the challenging domain of classic Atari 2600 games⁵. We demonstrate that the deep Q-network agent, receiving only the pixels and the game score as inputs, was able to surpass the performance of all previous algorithms and achieve a level comparable to that of a professional human games player across a set of 49 games, using the same algorithm, network architecture and hyperparameters. This work bridges the divide between deep reinforcement learning and games, resulting in the first artificial agent that is capable of learning to play at a diverse range of games.

We set out to create a single algorithm that would be able to develop a wide range of competencies in a varied range of challenging tasks—a central goal of general artificial intelligence⁶ that has eluded previous efforts^{7,8}. To achieve this, we developed a novel agent: a deep Q-network (DQN), which is able to combine reinforcement learning with a class of artificial neural networks known as deep neural networks. Notably, recent advances in deep neural networks⁹, in which several layers of nodes are used to build up progressively more abstract representations of the data, have made it possible for artificial neural networks to learn concepts such as object categories directly from raw sensory data. We use an particularly successful architecture, the deep convolutional neural network¹⁰, which uses hierarchical layers of local and non-local filters to extract the effects of receptive fields—inspired by Hubel and Wiesel's seminal work on fundamental processing in early visual cortex¹¹—thereby exploiting the local pixel-to-pixel correlations present in images, and building in robustness to natural transformations such as changes of viewpoint or scale.

We consider tasks in which the agent interacts with an environment through a sequence of observations, actions and rewards. The theoretical

agent is to select actions in a fashion that maximizes cumulative future reward. More formally, we use a deep convolutional neural network to approximate the optimal action-value function:

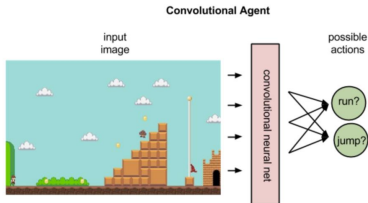
$$Q^*(s,a) = \max_{a'} \mathbb{E}[\sum_{t=0}^{\infty} \gamma^t r_{t+1} | s^0 = s, a^0 = a, s_1 = s', \dots, s_{t+1} = s', a_t = a, s_t = s']$$

which is the maximum sum of rewards discounted by a machine time step t , achievable by a behaviour policy $\pi = Q(a|s)$. After making an observation (s) and taking an action (a) (see Methods)¹², reinforcement learning is known to be unstable or even to diverge when a nonlinear function approximator such as a neural network is used to represent the action-value (also known as Q) function¹³. This instability has several causes: the correlations present in the sequence of observations, the fact that small updates to Q may significantly change the policy and therefore change the data distribution, and the correlation between the action values (Q) and the target values $r + \gamma \max_{a'} Q(s', a')$. We address these instabilities with a novel variant of Q-learning, which uses two key ideas. First, we used a biologically inspired mechanism termed experience replay¹⁴—the randomizes over the data, thereby removing correlations in the observations sequence and introducing new changes in the data distribution (see below for details). Second, we used an iterative update that adjusts the action values (Q) towards target values that are only periodically updated, thereby reducing correlation with the target.

While other stable methods exist for training neural Q networks¹⁵, these methods require the repeated training of networks of scores on hundreds of iterations. Consequently, these methods, unlike our algorithm, are too inefficient to be used successfully with large neural networks. We present an approximate value function, $\hat{Q}(s,a)$, using the deep convolutional neural network shown in Fig. 1, in which $\hat{Q}$ are the parameters that $\hat{Q}$ is a weights of the Q-network at iteration t . To perform experience replay we store the agent's experiences $\mathcal{D} = \{(s_t, a_t, r_t, s_{t+1})\}$ at each time step t in a list $\mathcal{D} = \{s_0, a_0, r_0, s_1, a_1, r_1, s_2, a_2, r_2, s_3, a_3, r_3, s_4, a_4, r_4, s_5, a_5, r_5, s_6, a_6, r_6, s_7, a_7, r_7, s_8, a_8, r_8, s_9, a_9, r_9, s_{10}, a_{10}, r_{10}, s_{11}, a_{11}, r_{11}, s_{12}, a_{12}, r_{12}, s_{13}, a_{13}, r_{13}, s_{14}, a_{14}, r_{14}, s_{15}, a_{15}, r_{15}, s_{16}, a_{16}, r_{16}, s_{17}, a_{17}, r_{17}, s_{18}, a_{18}, r_{18}, s_{19}, a_{19}, r_{19}, s_{20}, a_{20}, r_{20}, s_{21}, a_{21}, r_{21}, s_{22}, a_{22}, r_{22}, s_{23}, a_{23}, r_{23}, s_{24}, a_{24}, r_{24}, s_{25}, a_{25}, r_{25}, s_{26}, a_{26}, r_{26}, s_{27}, a_{27}, r_{27}, s_{28}, a_{28}, r_{28}, s_{29}, a_{29}, r_{29}, s_{30}, a_{30}, r_{30}, s_{31}, a_{31}, r_{31}, s_{32}, a_{32}, r_{32}, s_{33}, a_{33}, r_{33}, s_{34}, a_{34}, r_{34}, s_{35}, a_{35}, r_{35}, s_{36}, a_{36}, r_{36}, s_{37}, a_{37}, r_{37}, s_{38}, a_{38}, r_{38}, 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r_{600}, s_{601}, a_{601}, r_{601}, s_{60$

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